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19 April 1974
SUPERSEDING
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MILITARY SPECIFICATION

FABRIC AND TAPES, QUARTZ, IMPREGNATED WITH A HIGH FIXED CARBON CONTENT, MODIFIED PHENOLIC RESIN

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for quartz fabric impregnated with modified high fixed carbon content phenolic resin, and for tapes cut from the impregnated fabric.

1.2 Classification. Tape cut from the impregnated fabric shall be available in the following classes:

Class A, warp tape slit parallel to the warp direction of the fabric.

Class B, bias tape slit at an angle of 45 plus or minus 5 degrees to the warp direction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of the invitation for bids or request for porposal form a part of the specification to the extent specified herein.

STANDARDS

Federal

FED-STD-406 Plastics, Methods of Testing

FSC 8305

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Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or reuest for proposal shall apply.

American Society of Testing and Materials

ASTM D 579	Woven Glass Fabrics, Specifications and Tests For
ASTM D 1682	Breaking Load and Elongation of Textile Fabrics, Tests For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample is required and shall meet the requirements of this specification.

3.2 Material and dimensions. The material and dimensions shall be in accordance with the following:

3.2.1 Quartz cloth. The woven quartz shall be a 5 harness satin weave fabric made from yarns defined as quartz, continuous filaments, fiber diameter (.00035 inch nominal), 300 strand weight (yardage per pound, in hundreds). Abbreviated as QCG 300.

3.2.1.1 Finish. The fabric shall have an A - 1100 finish. The finish to be identified by the test method outlined in this specification.

3.2.2 Physical and mechanical properties. The dry fabric shall

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conform to the physical and mechanical properties listed in Table I of this specification.

3.2.2.1 Purity. The silicon dioxide (SiO_2) content of this fabric exclusive of binder shall be 99.95 percent minimum. The sum of sodium oxide (Ka_2O), potassium oxide (K_2O) and lithium oxide (Li_2O) shall not exceed 50 parts per million (50 ppm).

Table I. Physical and Mechanical Properties of Quartz Fabric

Property	Requirements
Weight-oz./sq. yd.	19.5 ± 2.0
Thickness-Inches	0.027 ± 0.004
Areal Shrinkage-%	1.0 max.
Breaking Strength-lb/in average	Warp-325 min. Fill-300 min.
Yarn Count per inch	Warp-38 ± 2 Fill-24 ± 2
pH	6.5 min. 7.5 max.

3.2.2.2 Additional requirements. Additional requirements such as bagginess, moisture content, workmanship, etc., may be specified by impregnator, providing there is not a conflict with other requirements of this specification.

3.2.3 Resin. The resin shall be a phenolic base resin with a fixed carbon content of 56 percent minimum

3.3 Impregnated fabric and tape requirements.

3.3.1 Dimensions. The minimum width of impregnated fabric shall be 36 inches. The dimensions of Class A and B material shall be as specified by the purchase order, except that tape width tolerances shall be plus or minus 1/16 inch and no more than one roll per width per shipment shall contain less than 40 lineal yards. Unless otherwise stated on the purchase order, no more than one roll per width per shipment shall be other than 17 plus or minus 1 inch outside diameter. The one roll per width per shipment may be less than 16 inches outside diameter.

3.3.2 Handling characteristics. Fabrics shall have sufficient best tack and drape to allow wrapping on compound surfaces. Tapes shall have the proper tack to permit machine wrapping of oriented configurations.

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3.3.3 Tape splices. Rolls of Class A tape shall contain not more than five (5) splices per roll. Class A or Class B tapes shall be joined using a sewn butt joint. The thread used shall be nylon or approved equivalent, and shall be kept uncontaminated during any sewing operation.

3.3.4 Storage. Classes A and B of materials when left unopened and sealed in the original containers shall have a minimum life of six (6) months when stored at 10 degrees Celsius (C) (50 degrees Fahrenheit (F)) maximum.

3.3.5 Physical. Classes A and B of materials shall be made from phenolic impregnation quartz fabric that conforms to the minimum physical and mechanical properties specified in Table II and Table III; except flow and volatile properties may be varied to suit molding or laminating processes. These exceptions shall not conflict with any other specified requirements.

Table II. Impregnated Fabric Requirements

Property	Requirement
Volatile Content (percent)	6 maximum
Resin Solids Content (percent)	30 plus or minus 3
Flow - (percent)	10 plus or minus 3

Table III. Properties of Cured Laminates

Property	Requirement (Minimum Average)	
	24 plus or minus 5.6 C (75 plus or minus 10 F)	260 plus or minus 5.6 C (500 plus or minus 10 F)
Tensile Strength (psi) (Warp Direction)	45,000	40,000
Tensile Modulus (psi) (Warp Direction)	2.5×10^6	2.0×10^6
Tensile Strength (psi) (Bias Direction)	12,000	10,000
Tensile Modulus (psi) (Bias Direction)	1.2×10^6	1.0×10^6

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Table III. Properties of Cured Laminates (Cont'd)

Property	Requirement (Minimum Average)	
Tensile Elongation (Percent) (Bias Direction)	1.0	1.0
Flexural Strength (psi) (Wrap direction)	48,000	22,000
Specific Gravity	1.7	---

3.3.6 Workmanship. Impregnated tape(s) shall be handled, processed, and stored in accordance with the best practices of the industry. They shall be uniform in quality, clean and free of tears, foreign matter, or other defects detrimental to fabrication or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent lots of material. Preproduction samples which do not meet all requirements of this specification shall be subject to rejection.

4.2.1 Preproduction sample size. Samples for preproduction tests shall be provided as follows:

a. Five (5) lineal yards of dry quartz fabric.

b. Forty (40) lineal yards minimum of phenolic impregnated quartz fabric (minimum of 4 inches wide) to meet Class A and B requirements.

c. Forty (40) lineal yards (minimum) of one inch wide (minimum), biased, phenolic impregnated quartz tape to meet Class A and B requirements.

4.2.2 Re-examination and retest. Re-examination of a product which has previously passed the preproduction tests shall be carried out with a preproduction sample, in accordance with this specification under any one of the following conditions.

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a. The manufacturer has modified his product or changed his material or processing sufficiently so that the validity of previous preproduction tests is questionable.

b. The requirements in the specification have been amended or revised sufficiently to affect the character of the product.

c. When deemed necessary to determine that the quality of the product is being maintained.

4.3 Classification of inspection. The examination and testing of the material shall be classified as follows:

a. Preproduction inspection.

b. Quality conformance inspection.

4.3.1 Preproduction inspection. Preproduction inspection shall be conducted only on the preproduction samples and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance inspection. Quality conformance inspection for acceptance of the impregnated fabric and tapes shall consist of the following examination and tests.

a. Resin solids content.

b. Volatiles.

c. Flow

d. Visual and dimensional inspection.

4.3.2.1 Lot size. For the purpose of sampling inspection and tests, a lot of the dry quartz fabric shall be defined as all material woven at one time from one batch of quartz yarn. A lot of preimpregnated fabric shall be defined as all material from one lot of fabric as defined above impregnated with one batch of resin. A lot of tape shall consist of all tape cut from one lot of preimpregnated material as defined above.

4.3.2.2 Batch. A batch of resin shall be defined as that quantity of resin processed through a mill or mixer at one time. A batch of yarn shall refer to that quantity of quartz yarns used in setting up the warp of the fabric during a single weaving process.

4.3.2.3 Sample size. For each type of material submitted, the

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supplier shall submit samples sufficient to obtain specimens from each sample as required by Figure 1 that are representative of the beginning of the lot, points within the run at intervals of 200 to 235 linear yards, and at the end of the lot during the impregnation of the broadgoods. Samples shall be broadgoods and the location of each sample relative to the portion of the run from which it was taken shall be furnished as a part of the identification of each sample. Tests shall be as noted in 4.3.2.

4.4 Certificate of conformance. The manufacturer shall present with each inspection lot a certificate of conformance stating that:

a. The dry quartz fabric meets the requirements of 3.2.

b. The resin used meets the requirements of 3.2.3.

NOTE: Certified test results from quartz fabric and resin manufacturers may be used for these materials in lieu of tests performed by the impregnator.

4.5 Test methods and examinations.

4.5.1 Dry quartz fabric (unimpregnated).

4.5.1.1 Purity. Emission spectrographic or chemical analysis methods shall be used to ensure that the material meets the requirements of 3.2.2.1.

4.5.1.2 Weight. The material shall meet the weight requirements of Table I and shall be tested in accordance with ASTM D 579, Section 7.

4.5.1.3 Thickness. The material shall meet the thickness requirements of Table I and shall be tested in accordance with ASTM D 579, Section 5 (10 psi).

4.5.1.4 Areal shrinkage. The material shall meet the areal shrinkage requirements of Table I and shall be tested as follows:

4.5.1.4.1 Sample preparation. Cut at least one 6 inch x 6 inch square specimen from each roll; identify and take at least three (3) measurements each of the length and width to the nearest .01 inch. Place the samples in a muffle furnace preheated to 982.2 plus or minus 28 degrees C (1800 plus or minus 50 degrees F) for 30 plus or minus 3 minutes. Remove and cool to room temperature and remeasure as before. In performing calculations, use the average of each dimension before and after firing. Area in both cases is equal to average length times average width.

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4.5.1.4.2 Calculation.

$$\text{Percent area shrinkage} = \frac{(\text{area before firing}) - (\text{area after firing})}{\text{area before firing}} \times 100$$

4.5.1.5 Breaking strength. The material shall meet the breaking strength requirements of Table I and shall be tested per ASTM D 579, except that covering the fabric with masking tape in the grip areas shall be an acceptable alternate procedure for the preparation of the test specimens.

4.5.1.6 Thread count. The material shall meet the thread count requirements of Table I and shall be tested per ASTM D 579, Section 8.

4.5.1.7 Hydrogen ion concentration (pH). The material shall meet the pH requirements of Table I and shall be tested as follows:

4.5.1.7.1 Sample preparation. Cut a sample of the quartz fabric from each lot of fabric into small pieces and weigh out one gram for testing. To not handle the sample with bare hands. Place the sample in a pyrex or polyethylene flank (250 milliliters (ml) size) and add 100 plus or minus 1 ml of distilled and freshly boiled water. Stir well and heat to 82.2 plus or minus 2.8 degrees C (180 plus or minus 5 degrees F) for 30 plus or minus 1 minute and allow to cool. Determine the pH of the water mixture using a pH meter. Measure the pH of the distilled water alone for reference purposes. Report both values.

4.5.1.8 Width and length. The material shall meet the width and length requirements of 3.3.1 and shall be tested by direct measurement.

4.5.1.9 Finish. Place a one inch square of fabric in a 50 cubic centimeter (cc) beaker. Add acetone to cover sample. Add water to the extent of five times the amount of acetone used. To this add 5 drops of a 10 percent solution of sodium nitro prusside. Rub the sample lightly with a stirring rod. A red color will appear in two or three minutes if A-1100 finish is present.

4.5.2 Impregnated quartz fabric.

4.5.2.1 Dimensions. The dimensions of Class A and B material shall conform to 3.3.

4.5.2.2 Volatile content. The volatile content shall meet the requirements of Table II. The test shall be run as follows:

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a. From each sample, cut three (3) specimens approximately 16 square inches and weigh each to the nearest milligram (.001 gram) in previously fired 871 plus or minus 28 degrees C (1600 plus or minus 50 degrees F) and tared porcelain crucibles. Record weight (W_1).

b. Dry in a circulating air oven for 15 plus or minus 1 minute at 162.7 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F).

c. Remove, cool to ambient temperature; reweigh to the nearest milligram and record (W_2). If more than one (1) hour is to elapse between removal from the oven and weighing, place the specimens in a desiccator.

d. Calculate volatile content as follows:

$$\text{Percent volatile content} = \frac{(W_1 - W_2)}{W_1} \times 100$$

Report individual values for each specimen and the average for all three specimens.

4.5.2.3 Resin solids content. The resin solids content shall meet the requirements of Table II and shall be run as follows:

4.5.2.3.1 Sample preparation. Place the dried specimens from 4.5.2.2 in a muffle furnace at 871 plus or minus 28 degrees C (1600 plus or minus 50 degrees F) for a minimum of 60 minutes or until a uniform white color and constant weight is obtained. Remove from furnace, place in a desiccator, cool to ambient temperature, and reweigh to the nearest milligram (W_3). Calculate resin solids content as follows:

4.5.2.3.2 Calculation.

$$\text{Percent resin solids content} = \frac{(W_2 - W_3}{W_2} \times 100$$

W_2 is defined in 4.5.2.2. Report individual values for each specimen and the average of all three (3) specimens.

4.5.2.4 Flow. Percent flow shall meet the requirements of Table II. The test shall be run as follows:

a. Cut four (4) inch x 4 inch square specimens on the bias from the impregnated fabric stack, weigh to the nearest milligram, and record (W_1).

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b. Cover the stack, top and bottom, with a glass-reinforced Teflon parting material and place in a press preheated to 162.7 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F) for 20 plus or minus 3 minutes at 150 plus or minus 10 psi. Press closure time shall not exceed five (5) seconds.

c. Remove from press, strip off parting material, scrape off all flash, taking care not to remove any fabric, and reweigh to the nearest milligram (W_2).

d. Calculate resin flow as follows:

$$\text{Percent flow} = \frac{(W_1 - W_2)}{W_1} \times 100$$

Repeat above test in triplicate and report individual values and the average of all three (3) determinations.

4.5.2.5 Mechanical properties test methods.

4.5.2.5.1 Test specimen preparation. Specimens for tests in Table III shall be prepared from a five ply laminate. The cure cycle shall be 85 plus or minus 5.6 degrees C (185 plus or minus 10 degrees F) at contact pressure for 25 plus or minus 3 minutes, followed by 162.8 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F) at 250 plus or minus 10 psi for 3 plus or minus .17 hours.

4.5.2.5.2 Tensile properties. Tensile properties shall meet the requirements of Table III and shall be determined in accordance with FED-STD-406, Method 1011. A minimum of five (5) specimens shall be tested for each property. Specimens shall be tested in the warp and bias direction of the fabric.

4.5.2.5.3 Ultimate flexural strength. Flexural strength shall meet the requirements of Table III and shall be determined in accordance with FED-STD-406, Method 1031. A minimum of five (5) specimens shall be tested. Specimens shall be tested in the warp direction of the fabric.

4.5.2.6 Fixed carbon content. The fixed carbon content of the resin shall meet the requirements of 3.2.3. The test shall be run as follows:

a. Place approximately three grams (3.0 plus or minus 0.5) of the resin solution as received from the manufacturer (approximately 60 percent resin solids) into a two-inch diameter aluminum dish for each specimen to be tested. If the resin does not spread to uniformly cover the bottom of the dish, it shall be spread with

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an appropriate clean spatula to insure a uniform thickness of the specimen.

b. Cure each specimen in a forced air circulating oven following the cure cycle below for oven times and temperatures:

51.6 plus or minus 2.8 degrees C (125 plus or minus 5 degrees F) for 2 plus or minus 0.1 hours, followed within 30 minutes by

65.5 plus or minus 2.8 degrees C (150 plus or minus 5 degrees F) for 4 plus or minus 0.1 hours, followed within 30 minutes by

93.3 plus or minus 2.8 degrees C (200 plus or minus 5 degrees F) for 4 plus or minus 0.1 hours, followed within 30 minutes by

121.1 plus or minus 5.6 degrees C (250 plus or minus 10 degrees F) for 4 plus or minus 0.1 hours, followed within 30 minutes by

148.9 plus or minus 5.6 degrees C (300 plus or minus 10 degrees F) for 4 plus or minus 0.1 hours.

c. Remove from hot (148.9 degrees C (300 degrees F)) oven and place in a desiccator until cooled to room temperature. Alternately, the specimen may be allowed to cool to room temperature in the oven and then be conditioned in a desiccator for a minimum of 24 hours before being removed from the dish.

d. Condition a 15 millimeter (mm) to 18mm diameter by 125mm to 150mm Vycor test tube for 30 minutes minimum in a forced air circulating oven maintained at 176.7 plus or minus 5.6 degrees C (350 plus or minus 10 degrees F). Remove the tube from the hot (176.7 degrees C (350 degrees F)) oven and place in desiccator to cool to room temperature. This test tube shall be left in the desiccator until ready for use. When ready for use, remove the test tube from the desiccator and weigh to the nearest 0.001 gram (W_1).

e. Remove the specimen to its dish from the desiccator and remove the resin from the dish. Carefully break the specimen into small flakes or chunks. Clean lint-free white gloves must be worn during this operation.

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f. Weigh out 0.8 to 1.0 gram of the resin chunks or flakes in the preconditioned Vycor test tube. Weigh the resin in the Vycor tube to the nearest 0.001 gram (W_2).

g. Place the Vycor test tube containing the specimen into a clamp mounted onto a ring stand. Connect vacuum hose to glass tube running through a neoprene or Viton stopper. Insert stopper into Vycor test tube to connect the test tube with a vacuum system that shall be set up as follows: Vacuum pump, approximately 18 inches of rubber vacuum tubing connected at one end to the vacuum pump and at the other end to a glass T-joint; from the T-joint, one length of rubber vacuum tubing approximately 12 inches long shall go to a vacuum manometer, and another length of rubber tubing approximately 12 inches long shall go to a cold finger trap; from the cold finger trap, a length of the rubber tubing approximately 12 inches long shall go to the glass tubing of the stopper inserted into the Vycor test tube.

h. Evacuate the system described in step g, above, to an absolute pressure of 1.5 mm of mercury or less. Check the system for leaks by clamping off the tubing leading to the vacuum pump for two minutes, minimum. Should the pressure rise by more than 2 mm during the two minutes, recheck the apparatus and seal all fittings with high-vacuum silicone grease.

i. At constant pressure, place the Fisher burner under the specimen in the Vycor tube, with the flame area completely covering the specimen. The pyrolysis temperature shall be a minimum of 800 degrees C (1470 degrees F) when checked with a pyrometer. The burner shall remain under the sample for 20 to 30 minutes. At the end of this pyrolysis period, the flame from the burner shall be moved so as to drive off as much condensate from the stopper end of of the tube as possible without damaging the stopper. At the end of the pyrolysis period the absolute pressure reading shall be less than 1.5 mm of mercury or else the specimen shall be discarded and another run in its place.

j. Shut off the Fisher burner and allow the specimen to cool to room temperature while maintaining an absolute pressure of 1.5 mm of mercury, maximum.

k. Shut off the vacuum, remove the Vycor tube from the vacuum system and place it in a desiccator to further cool for a minimum of two hours, or until constant weight is obtained. Weigh the tube and the pyrolyzed specimen to the nearest 0.001 gram (W_3). The tube shall be handled

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with clean tools, or if by hand, with the aid of clean, lint-free, white gloves only while being removed from the vacuum system, during transfer to or from the desiccator, or during weighing operations.

l. Calculate the fixed carbon content per the following formula:

$$\text{Percent Fixed Carbon Content} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \times 100$$

m. Test five (5) specimens and report the individual values and the average value. The average value shall be used for satisfying the 3.2.3 requirement, except that no single value shall fall below the required minimum value by more than 2.0 percent.

4.6 Shelf life. Six (6) months after the impregnation date, each package of impregnated tape which has been stored in accordance with the requirements of this specification shall be retested for resin flow and volatile content in accordance with 4.5.2.4 and 4.5.2.2 respectively. After sampling, the material shall be resealed in accordance with 5.3.1 and 5.3.2. Material meeting the requirements of this specification shall be kept in storage and those which fail on initial or subsequent reinspection shall be clearly identified as being unsatisfactory and shall not be used. This retest procedure shall be repeated every 90 days or prior to use, on material past the six (6) month storage requirement.

4.7 Workmanship. The tape(s) shall be visually examined for conformance to the requirements of 3.3.6.

4.9 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.2 Preservation. Material shall be shipped under conditions which will prevent deterioration in transit. Storage temperatures of sealed packages of material shall not exceed 10 degrees C (50 degrees F).

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5.3 Packaging Levels A and C. Tapes shall be wound on individual cardboard tube segments having a length the same as the width of the tape and an inside diameter of three inches. Individual layers of tape may be interleaved with .002 inch minimum thickness polyethylene. If interleaving material is not used, the tape layers shall not stick together when shipped and stored per 5.2. Individual rolls shall be heat sealed in a polyethylene bag of 0.002 inch minimum wall thickness.

5.4 Packing, Levels A, B, and C. Packaged material shall be shipped in commercial type containers suitably sealed to prevent contamination of the contents by moisture or foreign matter and shall have resealing features. Shipping containers shall be constructed to withstand normal transportation and handling abuse and shall be suitable for indoor storage for a period of 6 months minimum.

5.5 Marking. Shipping containers shall be marked in accordance with MIL-STD-129 and shall include date of packing.

5.5.1 Special markings. Unless otherwise specified (see 6.2), and in addition to the manufacturer's marking, a tag, or label shall be fastened on each cardboard roll of tape and a tag or label placed inside of or on each sealed bag containing tape or test samples bearing the following information:

a. Purchase order number.

b. Specification number and revision letter (if any).

c. Class of material.

d. Manufacturer's name and address.

e. Width of tape.

f. Net weight of material on roll.

g. Location of sample from lot of broadgoods (Acceptance Samples Only).

h. Lot number.

i. Date and place of impregnation.

j. Cloth lot number with manufacturer's designation.

k. Resin patch number with manufacturer's designation.

l. Storage requirements statement:

PERISHABLE: STORE AT OR BELOW 10 DEGREES C (50 DEGREES F)

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6. NOTES

6.1 Intended use. The material covered by this specification is to be molded or laminated by high pressure into ablative and insulation composities.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Applicable levels of preservation, packaging, and packing (see 5.1).

d. Sampling plan, if other than specified (see 4.3.1 and 4.3.2.3).

e. Class and cut required (see 1.2 and 3.3.1).

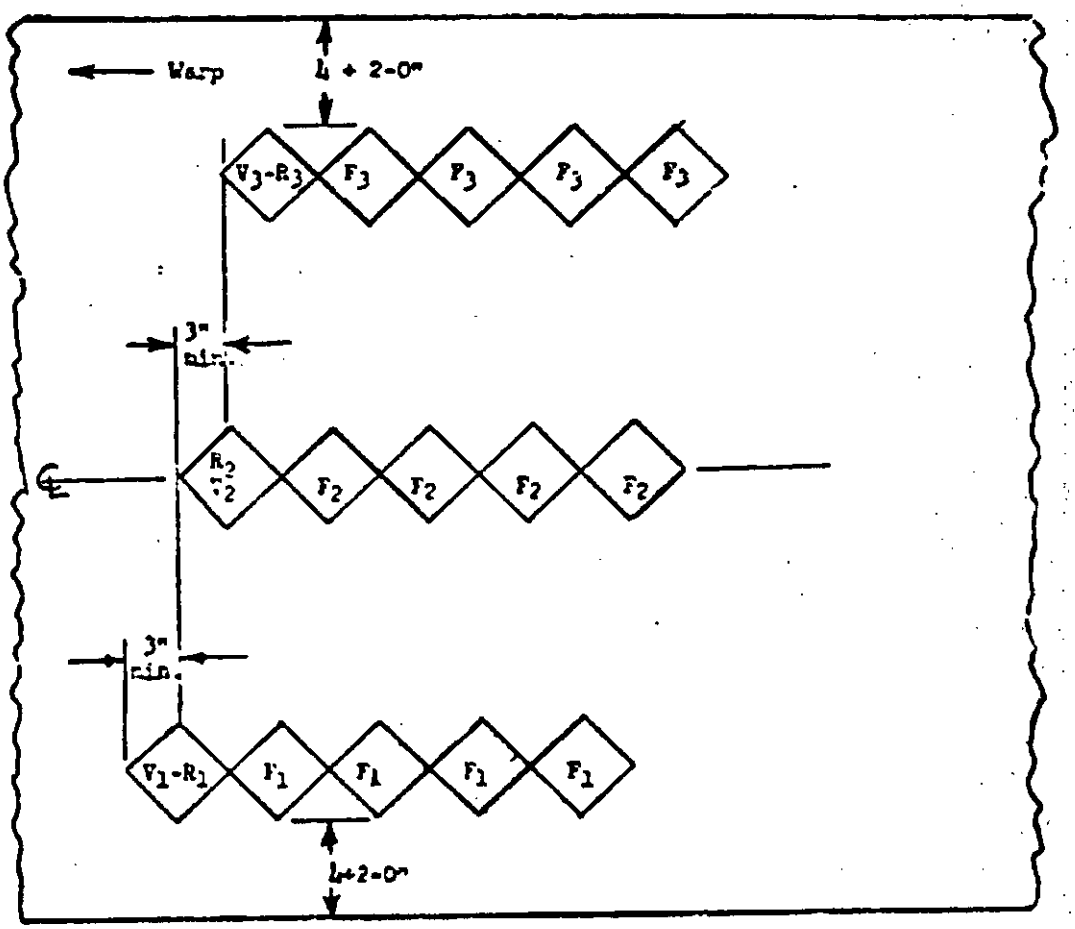
f. Width of Class A and B tape material.

6.3 Supersession data. This specification includes the requirements of MIS-13387A, dated 22 February 1972.

Custodian:
ARMY-MI

Preparing Activity:
ARMY-MI
Project No. 8305-A317

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V - R = Volatile and Resin Solids Content Specimens.

F = Flow Specimens

Figure 1.

Specimen sampling plan for each broadgood sample

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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